

Sediment recharge at Kellow Sands: the 2025/26 placement

Prepared by the technical secretariat for the Partnership Board · 12 February 2026

Background

The Partnership has placed dredged sand on the upper foreshore at Kellow Sands in each of the last three winters, under the standing consent granted in 2021 and the method agreed after the 2023 trial. The purpose is to slow the loss of intertidal in front of the saltmarsh, which the 2017 study put at rather more than a metre a year at the northern end.

The material is a by-product of maintenance dredging the Harbour Board carries out anyway. Placing it on the foreshore rather than at the licensed offshore site is cheaper for the Board and useful to the Partnership, and the arrangement has run without difficulty since 2021.

The consent position

The standing consent granted in 2021 runs to March 2029 and covers placement of up to 25,000 cubic metres a winter at the two locations shown in the consent drawing. It carries eleven conditions, of which two bear on the operation: placement is not permitted between 1 March and 31 August, and the Partnership must notify the regulator seven days before a campaign begins.

Both conditions were met this winter. The notification for the November start was sent on 28 October and acknowledged, and the January loads fell within the same campaign and needed no further notice. The regulator inspected once, on 4 December, and raised nothing.

The dredging programme

The material comes from the Harbour Board's maintenance dredging, which runs on a three-year cycle: the Vardenne Channel and the approach to the quay every year, the bar every second year, and the berths at Marnhead in the third. The volume available to the Partnership therefore varies with the cycle as well as with the winter, and the largest volumes fall in the years the bar is cut.

This was a channel-and-approach year, which is the smaller of the three. The Board's dredger worked twenty-two days in the estuary between October and January against thirty-one in the previous season, and the eight loads placed at Kellow Sands were what the programme could offer rather than what the Partnership asked for. The secretariat does not regard the difference as material at this stage of the trial.

The winter's placement

The Partnership placed 18,400 cubic metres of dredged sand on the upper foreshore at Kellow Sands between 6 November 2025 and 22 January 2026. The material came from maintenance dredging of the Vardenne Channel and was placed in eight loads under the standing consent granted in 2021. This note records what was placed, what it cost, and what the winter surveys

Placement followed the method agreed after the 2023 trial. Each load was discharged at high water on a rising spring tide, at a single point some forty metres seaward of the saltmarsh edge, and left to redistribute without machinery working on the flat. The intention is that the material moves landward under wave action over two or three winters rather than being spread mechanically, which the trial showed disturbs the buried peat horizon at Thrimby Point.

How the material is placed

Each load is brought up on the flood and held off the placement point until high water. Discharge takes between forty minutes and an hour and a quarter depending on the load, and the vessel is off the flat and back in the channel within the hour after. The Partnership's own staff attend every discharge from the shore and record the time, the state of the tide, the wind and anything of note; the records are held with the placement file.

The discharge point is fixed by two marks on the seawall and is the same point used in each of the three winters. Holding to it matters more than the volume: the whole intention of the method is that a single deposit redistributes under wave action, and a deposit spread over a wider footprint would not be the method the Partnership undertook to use.

Two discharges were stood down on the day for weather, one in November for wind and one in January for a swell running into the bay, and both were made good later in the same tide window. A stand down costs the Partnership nothing under the current rates, which is an arrangement the secretariat would like to keep in the fourth season and expects to have to argue for.

Volume and cost

The placed volume is close to the plan. One load was reduced because the dredger was released early to work at Marnhead, and the shortfall of roughly 600 cubic metres was not made up. Costs came to £164,300 against a budget of £172,000, of which £31,500 was the Partnership's own contribution and the balance was met from the coastal resilience allocation held by the Braithe Harbour Board.

The cost in detail

The figure of £164,300 is made up of four parts. Placement itself was £118,700, charged on the load rather than the cubic metre and therefore unaffected by the load that came in light. Mobilisation and demobilisation of the discharge gear was £21,400, which is a fixed cost and the reason a short season is proportionately dearer than a long one. The survey work, including the January resurvey and the laboratory's particle size analysis, was £16,900. The balance, £7,300, is the secretariat's own time at the rate agreed with the Board in 2023.

Against a budget of £172,000 that leaves £7,700 unspent, which the secretariat proposes be carried to the 2026/27 programme rather than returned, on the ground that the fourth season's

mobilisation charge will fall in a year when the dredging cycle offers less material. The Board's agreement to the carry-forward is one of the decisions sought below.

Two costs are not in the figure and should be understood. The Harbour Board's dredging is not charged to the Partnership at all: the material would be lifted whether or not it were placed at Kellow Sands, and only the difference in disposal is a cost to anybody. And the bird counts are made by the Vellamy Society without charge, which the Partnership has recorded before and records again here.

What the surveys show

Surveys have been run on the four fixed transects established in 2019. The January survey shows accretion on transects two and three, no measurable change on transect one, and a small loss on transect four, which sits closest to the training wall. None of these differences is yet outside the range recorded between winters before any placement took place, and the Partnership's advisers have asked that no conclusion be drawn before the third winter.

What the survey can and cannot show

The survey measures the profile of four fixed lines to a tolerance the method puts at fifteen millimetres, which is a good deal finer than the changes being looked for. Its limitation is not precision but coverage: four lines describe four lines, and the flat between them is inferred. A change concentrated between two transects would be invisible, and a change on a transect would be reported as though it were general.

That is a known limitation of the design rather than a fault in this winter's work, and it was accepted in 2019 as the price of a series that could be repeated cheaply every January. The alternative, a full topographic survey of the intertidal, was quoted in 2019 at eleven thousand pounds a visit and would not have been repeated annually. The Board may wish to revisit that at the end of the trial and not before.

The advisers' position

The Partnership's advisers were asked in December for a view on whether three winters of survey would support a conclusion. Their answer, which the secretariat accepts, is that it will not: the between-winter variation recorded before placement began is of the same order as the change measured since, and separating the two needs either a longer run or a control transect outside the placement footprint.

A control transect was considered in 2019 and rejected on cost. The advisers have suggested it be reconsidered, at an estimated four thousand pounds a year including the laboratory work, and the secretariat will bring that to the Board with the 2026/27 programme.

The peat horizon at Thrimby Point

The buried peat horizon at Thrimby Point is the reason the method is what it is. It is a designated feature, it lies under between a third and a metre of sand depending on the state of the foreshore, and the 2023 trial showed that a tracked machine working the flat disturbs it. The Partnership's undertaking not to work machinery on the flat was given to the regulator in writing and is a condition of nothing: it is a voluntary undertaking, which the secretariat mentions because voluntary undertakings are the easiest thing to forget.

The horizon is inspected each spring at three points by the Partnership's ecologist, and the 2025 inspection found no exposure and no evidence of disturbance. The 2026 inspection is due in April. Nothing in the placement method should affect it, and the inspection exists to establish that rather than to look for a problem anybody expects.

Water quality during placement

Turbidity in the creek at Vardenne rose after each discharge and fell back within a working day, which is the pattern of the previous two winters and was expected. Suspended solids at the Vardenne Quay station were reported at 41.6 milligrams a litre two days after the last placement, against figures below thirty for December and January. The laboratory's note attributes the difference to the placement and the state of the tide, and the secretariat accepts that.

No other determination moved outside its usual range, and the programme's quarterly summary will report the winter as unremarkable on water quality. The Partnership has not been asked for turbidity monitoring during discharge and does not propose to offer it: the creek is not a designated water, the effect is short-lived, and the cost would come out of the survey budget, which is the part of the programme that answers the question the trial is asking.

Birds and vegetation

Vegetation on the marsh edge was mapped in October, before placement, on the line used since 2019. The mapped edge has moved seaward by a little under two metres at its northern end and is unchanged elsewhere, which is within the variation the ecologist records between years and is not offered as an effect of the recharge. The 1998 edge, which the advisers use as the reference, is some eleven metres landward of the present one at the same point.

No change was recorded in the extent of the pioneer zone, which is the part of the marsh a sediment supply would be expected to affect first and the part the Partnership watches most closely. The ecologist's view, recorded in her October note, is that three winters is too short a period to expect a vegetation response at all.

Matters raised locally

Two matters were raised by residents during the placement period. The first concerned lorry movements on Kellow Lane, which were outside the hours set in the consent on three days in December; the contractor has accepted this and the hours have been restated in the site

instructions. The second concerned turbidity in the creek at Vardenne, which was visible for several hours after each discharge and is expected, but which had not been explained locally in advance.

The bird counts made during the placement window show no change in the numbers of wintering dunlin and grey plover on the flats north of the placement point. Counts were made by the Vellamy Society under its usual arrangement with the Partnership and are reported separately. The Society has asked whether placement can be kept outside the January count period in future years, and the secretariat sees no operational reason why it cannot.

Talking to people beforehand

The Partnership held a drop-in session at the Assembly Rooms on 18 October, before the season began, at which the programme, the consent conditions and the lorry route were explained. Twenty-three people came, most of them from Kellow Lane, and the note of the session is on the file. The two matters raised in December had both been raised at the session, which the secretariat takes as a sign that the session worked and that the follow-through did not.

For the fourth season the secretariat proposes a letter to every household on Kellow Lane a fortnight before the first load, giving the dates, the hours and a telephone number that is answered. That is what was missing this winter: the information existed and the route to it did not, and a resident who has to find the secretariat to ask about turbidity in their own creek has been let down whatever the answer turns out to be.

The contract and the contractor

The contractor performed the placement to the specification and without incident on the water. The failure was on land: lorry movements on Kellow Lane outside the hours set in the consent on three days in December, which the contractor has accepted in writing. The site instructions have been reissued with the hours in bold on the first page, and the Partnership's site staff have been asked to record the first and last movement of every day whether or not anything appears wrong.

The final account has been received and is £3,100 above the interim valuation, on account of the two weather stand-downs and a claim for standing time on 14 January which the secretariat has rejected. The account is expected to settle at or near the interim figure and within the budget reported above. The Board will be told if it does not.

The trial method, three winters on

The 2023 trial compared mechanical spreading against single-point discharge on two transects, and the finding that settled the method was not about sediment at all: spreading disturbed the buried peat horizon at Thrimby Point, which is a designated feature, and the Partnership undertook not to work machinery on the flat again. Single-point discharge was the alternative rather than the preference.

Three winters on, the case for the method is still the peat rather than the sediment. The advisers'

view is that mechanical spreading would probably move material landward faster. The Partnership is not able to do it, and the question does not arise unless the undertaking is revisited.

Risks

Two risks are worth recording. The first is the dredging contract: if the Harbour Board's retender in June produces a contractor without a suitable dredger, the material may not be available and the winter's placement would not happen. The second is the peat undertaking, which constrains the method to single-point discharge and would have to be revisited if the surveys eventually show that the method does not work.

How the programme is governed

The programme is authorised by the Board annually and delivered by the secretariat under the standing consent. The regulator is notified before each campaign and inspects at its own discretion; the Harbour Board is a party to the arrangement for the material and receives the quarterly report; and the Partnership's advisers are engaged for the survey and for an annual opinion on what the series will support.

What the arrangement lacks is a point at which the trial is formally reviewed against a decision. The secretariat proposes that the fifth winter be that point, that the review be commissioned in the autumn of 2027 rather than assembled from the annual reports, and that the Board agree the questions the review is to answer at the same meeting as it agrees the fourth season. A trial without a decision at the end of it becomes a programme by default, which is not what was agreed in 2021.

Options for the fourth winter

Four courses are open for the fourth winter. The first is to continue as at present, which costs what this winter cost, adds a fourth year to the series and answers nothing new. The second is to continue and add a control transect outside the placement footprint, at an estimated four thousand pounds a year, which is the advisers' recommendation and the only one of the four that improves what the series can show.

The third is to pause placement for a winter while continuing the survey, which would give a year of recovery data and would probably end the arrangement with the Harbour Board, whose dredger cannot be held for a season that may not happen. The fourth is to stop, which the secretariat does not recommend and sets out only because a trial that cannot be stopped was never a trial.

The secretariat's view is the second course. The marginal cost is small against the programme, the control transect answers the advisers' objection to the design rather than merely adding to the run, and it can be established in the same January window by the same crew.

The financial outlook

The coastal resilience allocation held by the Harbour Board runs to March 2029 and has £340,000 uncommitted against it. On the present rate of spend, and allowing for the control transect and for inflation on the dredging contract, the allocation covers the trial to its conclusion with about a season's contingency in hand. It does not cover a programme of works if the trial concludes in favour of one, and the Partnership should not proceed as though it did.

The dredging contract is retendered in June and is the largest uncertainty in the figure. The Board's own estimate is an increase of between eight and fourteen per cent, which on this winter's placement charge would be between nine and seventeen thousand pounds. The secretariat will report the outcome with the September programme paper.

Work remaining this winter

The work remaining this winter is limited to the February profile survey, which was postponed by weather and is now expected in the week of 23 February, and the reconciliation of the contractor's final account. Neither is expected to change the position set out above.

Next winter

The 2026/27 programme will be put to the Board in September with the contractor's rates for a fourth season. The secretariat expects the volume to be similar and the cost to rise with the dredging contract, which is retendered in June.

Recommendation

The Board is asked to note this record of the winter's placement, to agree that the January count period is avoided in the 2026/27 programme, and to confirm that the third winter of survey should be completed before the method is reviewed.

Enquiries to the technical secretariat, Braithe Estuary Partnership, Estuary House, Vardenne Quay, Braithe BR3 4LN.